

Investigating the Fidelity of Digital Peer Support: A Preliminary Approach Using Natural Language Processing

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Introduction

Peer Support can improve recovery outcomes such as decreased hospitalization, better quality of life, and fewer anxiety/depression symptoms.

Feasibility, acceptability, and preliminary effectiveness of digital peer support has been demonstrated in various studies.

Despite unique challenges, no validated measure to assess the quality and fidelity of peer support exists.

Aim: To understand the individual components of a digital peer support session, establish measures of high and low fidelity, and finally build a classifier to identify these indicators in novel datasets.

H0: The fidelity classifier will not be able to identify high and low fidelity phrases in a novel dataset.

H1: The NLP model will achieve a level of accuracy defined as 70% when classifying between high and low fidelity texts.

Methods

Data: 27 PeerTECH audio-to-text transcripts + comments from 6 subreddits and 6 Facebook support groups

Extraction and Coding

- extensible Human Oracle Suite of Tools
- Python Reddit API Wrapper

Dictionary

- Setting up Classes and Attributes
- Grounded Theory Approach

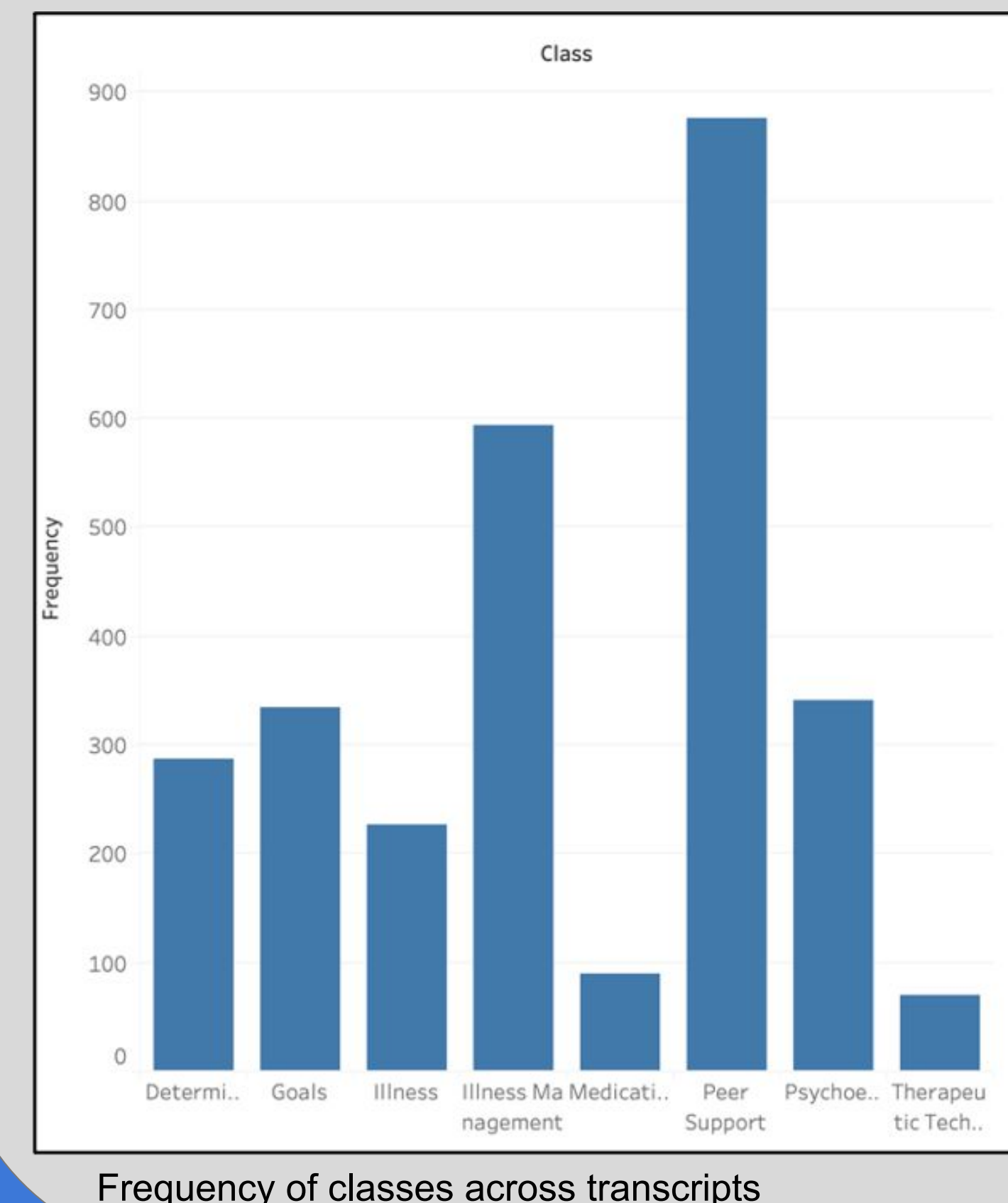
Fidelity Classification

- Combining fidelity measures and CPS manuals
- Binary classifying
- CPS validation

Topic Modeling & Classifier

- Latent Dirichlet Allocation
- BERTopic
- DistilBERT

Data Dictionary and Fidelity Measure



Data dictionary constructed with 8 classes.

Most frequent code (and corresponding attribute) for each class:

Determinants: family (protective factor: family)

Goals: healthier (illness recovery goals)

Illness: depression (diagnosis)

Illness Management: medication (relapse prevention: medication use)

Medication: weight gain (side effects)

Peer Support: that's okay (validation)

Psychoeducation: mind and body are connected (physical, social, and mental health connection)

Therapeutic Techniques: positive self-talk (cognitive)

Fidelity categories - **Process, Attitude, Content, and Techniques**

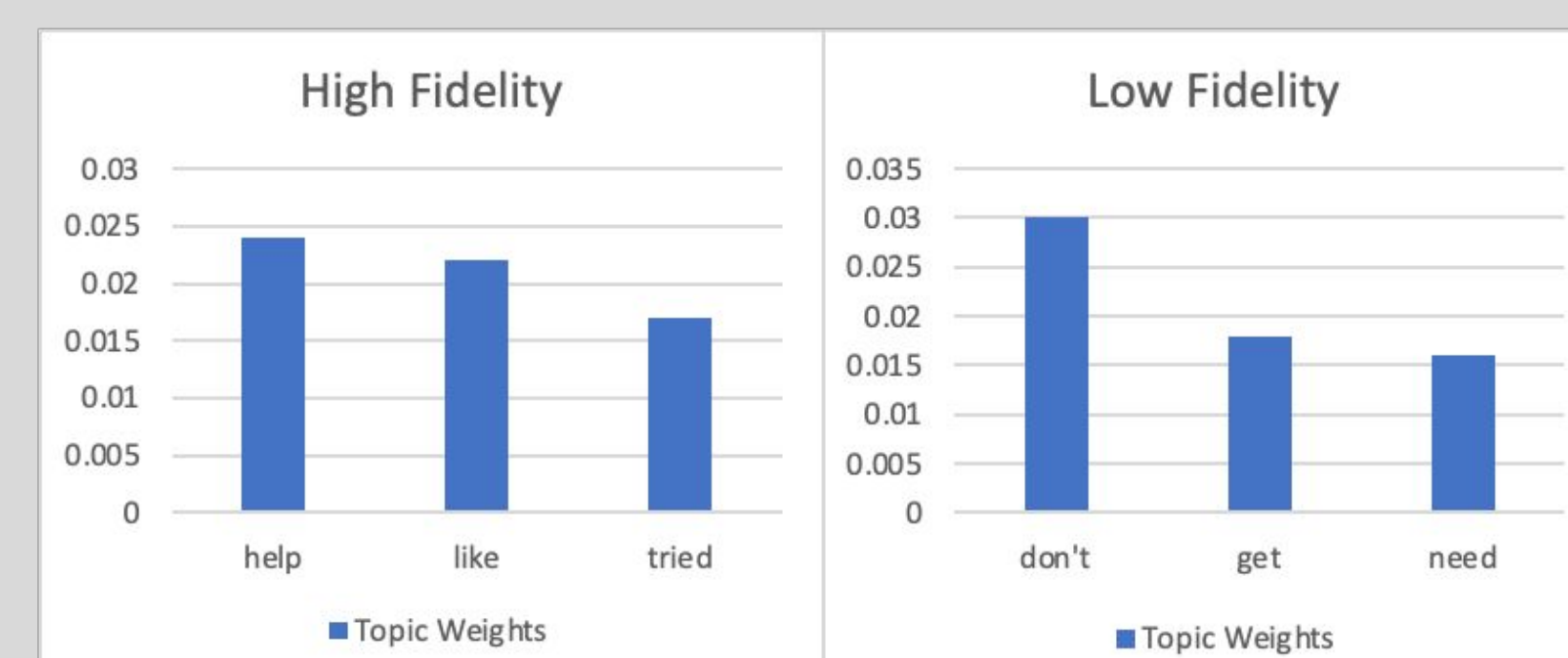
Overlap between **Illness Management, Peer Support, Psychoeducation,** and **Goals** classes with **Techniques** and **Content** fidelity categories.

BERTopic and LDA

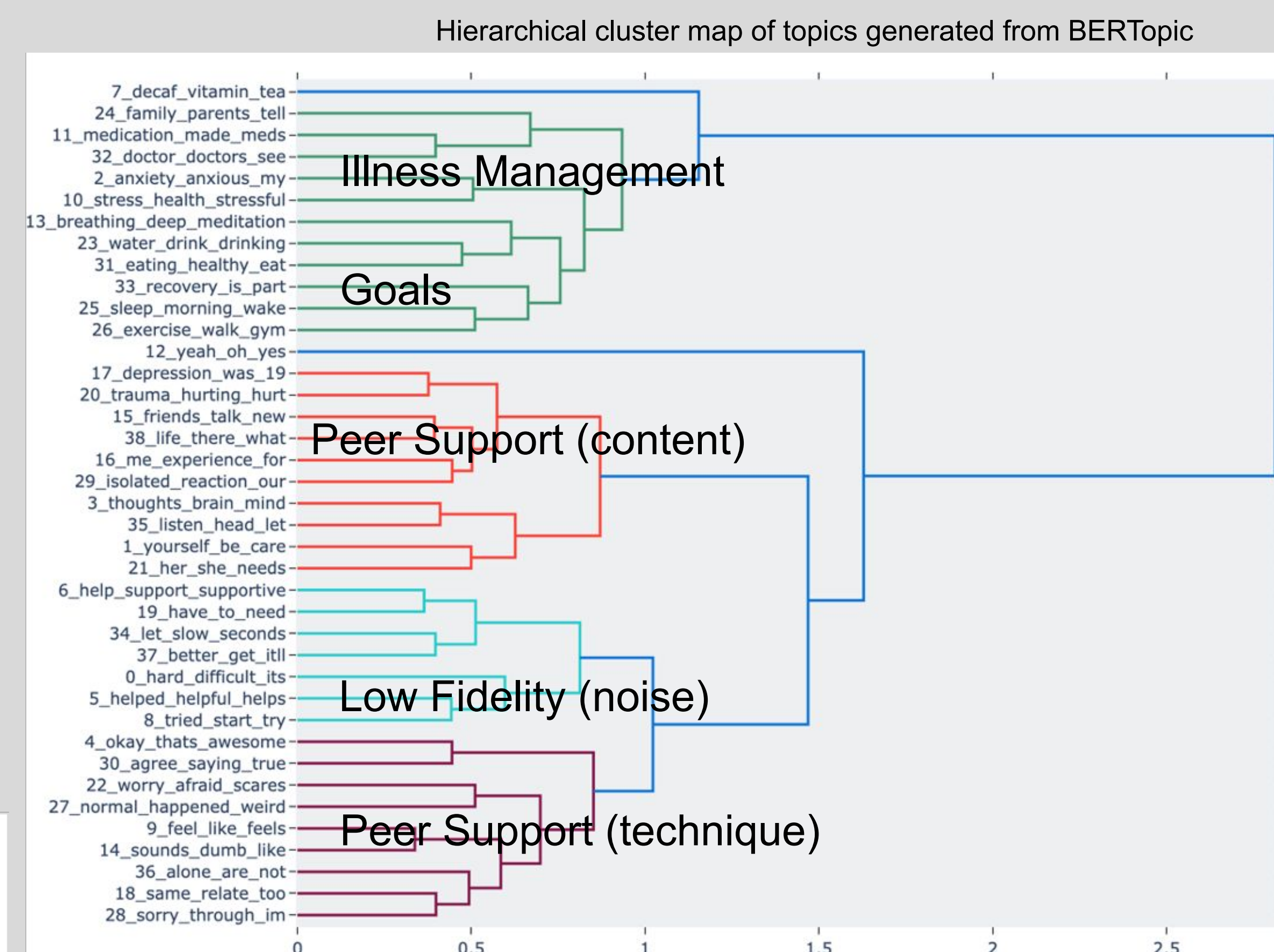
29/40 meaningful topics

Clusters overlap with same classes that were evident in fidelity measure.

Topics developed represent individual attributes.



Words with highest topic weights generated by LDA



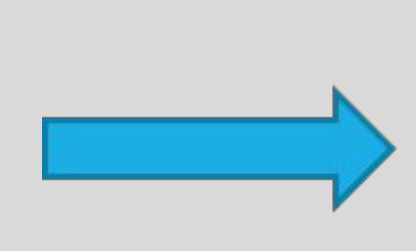
For two topics, LDA generated low coherence scores (-15.5 for low fidelity, -14.1 for high fidelity).

Non-significant Mann-Whitney U test ($U = 43$, $p > .05$) suggests that both categories had similar distributions of weights for its words.

DistilBERT



DISTILBERT INITIAL TRAINING RUN



- ACCURACY: 76%



- LOSS: 0.66

Future Scope

Find correlation between the outcomes of NLP algorithms and evidence-based medical, psychiatric, and social health manuals.

Controlled experiments to study feasibility and effectiveness.

Increase dimensions – measuring fidelity through speech (pauses, response time, etc.).

Assign scores indicating the degree of fidelity to transcripts of conversations.

Automatically flag texts or speech when CPS is deviating from fidelity adherence.

Provide real-time auditing, supervision, and feedback.

Conclusion

Methodology and classifier can be adapted for other kinds of telehealth.

Benefits task shifting – reduces clinician load.

Chance to increase the self-determination of peer support specialists.

Fidelity scores can be referenced with other mHealth resources (e.g., behavioral sensing, EMAs) to monitor service users.

Generate reports on quality metrics and key features of sessions.